DATASCI 406 FA 26, Quiz 1.4

Name and Uniqname:

Question 1 (1 point)

You simulate $B = 999$ datasets under H_0 and find that **every** simulated value of the test statistic is smaller than the observed t_{obs}. What do you report? **Circle one.**

1. $\hat{p} = 0$
2. $\hat{p} = 1/1000$
3. $\hat{p} = 1/999$
4. $\hat{p} < 0.05$; the simulation cannot pin the value down any further.

Solution

Answer: 2. $\hat{p} = (1 + \#\{T_b \geq t_{\text{obs}}\})/(B + 1)$ counts the observed statistic among the draws. It can never return 0, and it makes the test exactly valid rather than approximately so. Option 1 calls the data impossible under H_0 when we only failed to simulate enough; option 3 has the right idea and the wrong denominator.

Question 2 (1 point)

$X_1, \ldots, X_{25}$ are independent draws from $N(\mu, 1)$; the statistic T is the **sample median**; $H_0 : \mu = 0$ against $H_1 : \mu = 0.35$; and the rejection region was fixed in advance as $T > 0.3$.

```
set.seed(406)
null <- replicate(20000, median(rnorm(25, 0.00, 1)))
alt  <- replicate(20000, median(rnorm(25, 0.35, 1)))
round(c(A = mean(alt),      B = mean(null > 0.3),
        C = mean(alt < 0.3), D = mean(alt > 0.3)), 3)
```

```
##     A     B     C     D
## 0.352 0.116 0.420 0.580
```

Which printed value is the **size** of this test, and which is its **power** against $\mu = 0.35$? Write one label in each blank.

size = _____ power = _____

Solution

size = `B` (0.116), power = `D` (0.580).

Both average the same indicator, $T > 0.3$. They differ only in which world made the data: `null` under H_0, `alt` under H_1. Read the error table down a column.

- `C` $= 0.420$ is β — the power's world, counting the failures ($0.420 + 0.580 = 1$).
- `A` $= 0.352$ averages the **statistic**, not an indicator. It estimates the alternative's median: a fact about the data, not about the test.

The size is 0.116, not 0.05. The cutoff was fixed first, so the size is whatever it delivers. Resize with `quantile(null, 0.95)` $= 0.408$ and the power falls to 0.41 — which is why power only compares at a common size.

Rubric. 1 pt for both blanks; no partial credit.

Question 3 (3 points)

$X_1, \ldots, X_{40}$ are independent Exp(rate $= \theta$) with θ unknown. The test statistic is the 25% trimmed mean `mean(x, trim = 0.25)`, which has no convenient sampling distribution. The observed data give $t_{\text{obs}} = 1.39$.

Suppose we have implemented a **two-sided level-0.05 test** of $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$, for one particular θ_0 using a Monte Carlo hypothesis test. The procedure is implemented as a function `test(t_obs, theta_0)` that returns `TRUE` if the hypothesis is rejected and `FALSE` if the hypothesis is not rejected.

Using this function, describe how you would produce a **95% confidence interval for** θ. Your answer must say both (i) what you would compute, in enough detail that someone could code it from your description, and (ii) why what you get is a 95% interval. You may suppose that θ is known to fall in some range $[a, b]$.

Solution

(i) Hold the data fixed, so `t_obs` never changes, and call `test()` at every candidate θ_0 across $[a, b]$. Keep the ones it does **not** reject; that set is the interval.

```
grid <- seq(a, b, by = 0.005)
reject <- map_lgl(grid, function(th) test(t_obs, th))
range(grid[!reject])
```

One point for the sweep, one point for keeping values where function returns FALSE.

(ii) Let θ be the true value. The test has level 0.05, so handed the true null it returns `TRUE` with probability 0.05. The kept set is exactly where it returned `FALSE`, so it contains θ with probability 0.95. Therefore it is a proper 95% CI.

No credit for restating how the test works in place of a coverage argument.

One point for correct justification.